

Multi-Model Demand Intelligence Built for BUILT Finance

Mo is the BUILT-branded Aevah interface that runs multiple machine learning predictive models together for each analytic: demand forecasting, cannibalization detection, price elasticity, causal promo readouts, and explainability. It turns work that would otherwise require separate spreadsheets, statistical tools, BI workflows, and analyst review cycles into one continuously improving planning system grounded in live SPINS data.

250-535

analyst hours
replaced per four-
week refresh

4.4%

best Q1 2026 SKU x
retailer wMAPE
result

63%

price-direction
accuracy on clean
price events

\$22M+

annual value case at
\$1M per one-point
improvement

LEAD FACTS

Mo Runs a Multi-Model Forecasting System Behind One BUILT Interface

Mo does not rely on one forecast formula. For each analytic, it coordinates the right machine learning model stack: quantile demand forecasts for floor/plan/ceiling, causal counterfactuals for price and promotion events, retailer-specific elasticity, rolling SKU-to-SKU cannibalization pressure, and SHAP driver explanations. The finance value is not just better math; it is the ability to run, refresh, compare, explain, and audit those outputs every SPINS cycle without rebuilding the workflow by hand.

Executive takeaway: Mo compresses a multi-workstream forecasting operation into a repeatable finance workflow. The value is not a dashboard; it is a faster operating plan, cleaner trade spend economics, SKU-level portfolio visibility, and a decision trail your team can defend in a forecast meeting.

Demand Planning Analytic

LightGBM quantile forecasting, rolling validation, and SHAP explanations produce P10/P50/P90 demand ranges for SKU x retailer planning.

Trade Spend Analytic

Retailer elasticity, BSTS counterfactuals, and promo mechanic decomposition separate price-only lift from display, feature, and seasonal effects.

Assortment Analytic

8-week SKU-to-SKU pressure signals, distribution controls, and launch lifecycle context distinguish additive demand from internal share transfer.

Manual / Legacy Effort Equivalent

The estimate below uses the current evidence base in this brief: 164 active SKU x retailer series, a four-week SPINS refresh cycle, P10/P50/P90 planning ranges, retailer-level price sensitivity, event-level causal readouts, cannibalization pressure, and driver explanations. It is a planning estimate for comparable manual or legacy workflows, not a claim that BUILT's team currently spends this many hours.

Capability Mo Automates	Manual or Legacy Equivalent	Effort Estimate
13-week SKU x retailer demand ranges	Build and QA floor/plan/ceiling forecasts for 164 active series using spreadsheet, BI, or statistical-tool workflows.	~55-80 analyst hours per refresh, assuming 20-30 minutes per series.
Four-week retraining and validation	Rebuild features, refresh model inputs, rerun holdout checks, compare fresh vs. stale performance, and reconcile exceptions.	~40-80 hours per SPINS cycle before management review.
Price and promo counterfactuals	Create event workbooks, choose control periods or control retailers, isolate price from display, feature, seasonality, and distribution.	~2-4 hours per event; 10 events becomes ~20-40 hours.
Retailer-level elasticity	Maintain account-specific price-response studies with controls for distribution, maturity, pack size, promotion, and seasonality.	~8-16 hours per key retailer; eight retailers becomes ~64-128 hours.
Cannibalization pressure	Manually compare co-stocked SKU pairs by retailer over rolling 8-week windows, then separate substitution from distribution and promo effects.	~40-165 hours per review, depending on whether 500 or 1,000 SKU-pair checks are reviewed at 5-10 minutes each.
Driver explanations and audit trail	Document why each forecast moved, tie it back to source data, and prepare finance-readable explanations for forecast meetings.	~27-41 hours per refresh, assuming 10-15 minutes across 164 series.

Bottom line: Recreating Mo's current capability with manual processes or legacy point tools would reasonably require roughly **250-535 analyst hours per four-week refresh** if the workstreams above were handled separately. That is about **1.6-3.3 full-time analyst months per SPINS cycle**, excluding the initial data engineering, model operations, version control, and dashboard buildout needed to make the process repeatable.

SPINS DATA ALONE

13.1%

Weighted forecast error achievable from SPINS POS data — 5-quarter rolling average

BUILT'S CURRENT PROCESS

7-10%

Corporate-level accuracy — a strong baseline aggregating across accounts and SKUs

MO UNIFIED INTELLIGENCE

4.4%

Aevah unifies all data sources with blended machine learning — best Q1 2026 result

ELASTICITY VALIDATION

63%

Direction accuracy on clean, non-confounded price events

VALUE CASE

\$22M+

Annual value at \$1M per 1pp improvement (portfolio avg)

The Forecasting Foundation

SPINS Point-of-Sale data alone — using statistical methods — can get weighted forecast error (wMAPE) to around 13.1% across the portfolio. BUILT's current process, aggregating data across accounts and SKUs, achieves a strong 7–10% at the corporate level. Mo goes further: by unifying all available data sources with a blended machine learning ensemble, Aevah reached **4.4% wMAPE** in Q1 2026 on a 13-week hold-out across 164 active series. The goal is to continuously extend that precision down to individual SKU × retailer detail — compounding accuracy with every four-week retraining cycle.

What continuous retraining delivers: Every four-week SPINS cycle, Mo retrains on the latest data — so accuracy reflects the current portfolio, not a model built six months ago. In testing, a stale model degraded nearly 18 percentage points versus a freshly retrained one. Your team gets a forecast signal that keeps pace with BUILT's rapid SKU launches, retailer resets, and distribution changes — automatically, without additional analyst effort.

13-Week Retailer Forecast — Floor · Plan · Ceiling

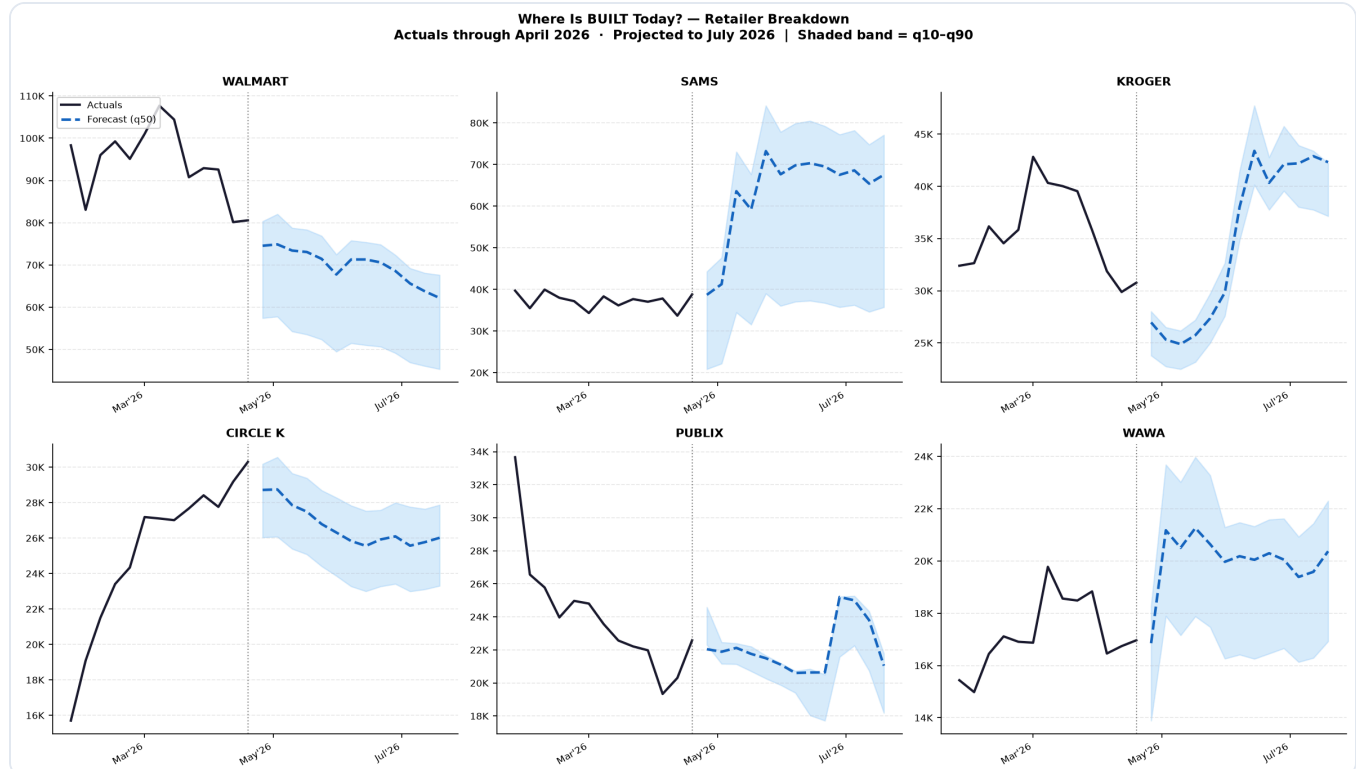


Figure 1 — Retailer-level 13-week forward projection (top 6 accounts by volume). Dark line = last known SPINS week. Shaded band = q10–q90 demand range. Dashed center = q50 operating plan. Each panel is a separate model series — no aggregation blending.

Supply chain payoff: Three numbers per account, not one. Floor drives safety stock. Ceiling drives capacity reservation. Center drives the operating plan. When a single number is wrong, the entire inventory position is wrong. A calibrated range lets your team plan around realistic upside and downside simultaneously.

A Continuously Improving Ensemble

What is an ensemble model? An ensemble combines multiple specialized models — each optimized for a different aspect of the forecasting problem — into one more accurate signal. Rather than relying on any single approach, the ensemble routes each forecast series through the methods that work best for that type of series and blends the results. The same principle applies in financial forecasting: combining multiple well-reasoned perspectives consistently outperforms any single forecast alone.

Mo's ensemble uses three layers working together: Light Gradient Boosting Machine (LightGBM) handles the core demand signal using 27+ SPINS-derived features. Bayesian Structural Time Series / CausalImpact (BSTS) isolates the causal effect of specific price and promotion events. SHapley Additive exPlanations (SHAP) translates every prediction into a ranked list of business drivers in plain English. The ensemble retrains every four weeks as new SPINS data arrives — matching your own planning cycle — so accuracy compounds rather than decays.

Model Layer	What It Contributes	FP&A Benefit
Light Gradient Boosting Machine — LightGBM (quantile)	P10/P50/P90 demand forecast using distribution, velocity, price, promo, seasonality	Calibrated floor/plan/ceiling for every SKU x retailer
Bayesian Structural Time Series / CausalImpact (BSTS)	Bayesian counterfactual: what would demand have been without a specific price or promo event?	Validates trade spend ROI with causal evidence, not correlation
SHapley Additive exPlanations (SHAP)	Ranks the contribution of every feature to every forecast — in plain-English business terms	Every prediction has an auditable paper trail your finance team can inspect
4-week retraining	Rolling window retrain each SPINS cycle; stale model degrades ~18pp without it	Accuracy stays current as SKUs launch, retailers reset, and your portfolio grows

Cannibalization Detection

BUILT expanded from roughly 3.5 SKUs to 7.5 SKUs in a year and is still growing. Every new launch creates the same FP&A question: *are those units incremental demand, or are they transferring volume from an existing BUILT product?* Answering it wrong leads to overstated forecasts, excess inventory commitments, and trade spend on launches that are eating your own portfolio. Mo tracks 8-week rolling SKU-to-SKU pressure signals across the entire portfolio — giving your team an early read before the miss shows up in a quarterly revenue call.

What Mo measures: 8-week Pearson correlation between co-stocked BUILT SKUs at the same retailer. A consistently negative co-movement signal — one SKU rising as another falls — is flagged as cannibalization pressure. Strength, direction, and confidence evolve weekly as the sales pattern matures.

Where Cannibalization Shows Up in Your FP&A Work

FP&A Scenario	Without Mo	With Mo
New flavor launch forecast	Estimate incremental units by analogy or by comp SKU; cannibalization is assumed but not measured	Mo tracks cross-SKU pressure from week 8 onward; signals whether new units are additive or transferred
Pack size expansion (1ct → 4pk)	Compare velocity manually; hard to separate channel mix from true demand substitution	Mo controls for distribution (Total Distribution Points, or TDP) and price, isolating the demand transfer signal at the same retailer
Revenue quality review	Portfolio growth looks good in aggregate; internal share shifts are invisible until the quarter closes	SKU-level portfolio pressure visible in real time; distinguishes earned growth from intra-brand substitution
Inventory commitment for Q3 launch	Commit to full build on optimistic incremental assumption; learn the transfer rate retrospectively	Early cannibalization signal informs the commitment range before the inventory order is placed

So what? When the Q3 new SKU launch is in the operating plan, Mo gives your team the signal to distinguish additive demand from internal share shift — before you commit to inventory and trade spend. With the launch calendar in the model, cannibalization pressure can be tracked from launch week 8 onward.

Price Elasticity & Trade Spend Intelligence

Not every retailer responds to price the same way. Mo measures per-retailer price sensitivity from real SPINS transaction history — controlling for distribution, promotion, seasonality, and pack size lifecycle — so your team knows where a price move will move units and where it will not. This is the input trade teams have always needed for promo ROI decisions but have never had at account-level precision.

Per-Retailer Price Sensitivity (Confirmed Negative Elasticity)

Retailer	Elasticity (ϵ)	Plain-English Read	Trade Spend Implication
Walmart	-0.245	Inelastic — shoppers not highly price-sensitive at this account	Small Temporary Price Reduction (TPR) moves carry limited volume upside; spend may be better allocated elsewhere
Kroger	-0.590	Moderate — price moves are felt; validated by December 2025 event	Promo return-on-investment calculations should use $\epsilon = -0.59$; causal lift confirmed at +28.6% with full promo mechanics
Ahold Delhaize	-1.262	Elastic — demand responds strongly to price changes	Price moves require demand offset planning; strong ROI opportunity when promo conditions are right
Whole Foods	-0.445	Muted sensitivity — consistent with premium positioning	Price response is lower than mass/grocery; velocity and distribution are stronger levers here

The Kroger Case — Causal Price Event Analysis

In December 2025, Brownie Batter 4pk at Kroger dropped from \$10.99 to \$10.14 (-7.7%). Mo's BSTS counterfactual model isolated the price-only effect from everything else happening that period — broader seasonal uplift, display and feature support, distribution changes — and answered: *how many units were genuinely above the counterfactual baseline because of the price change?* The answer: price alone explained +4.7% lift. The full promo package (price + display + feature) drove +28.6%. The gap is the promo mechanic value — quantified for the first time at event level.

CausalImpact Event — BB 4pk at Kroger: Dec 2025 Price Reduction
Pre-period May-Nov 2025 · Intervention Dec 7 2025 · ARP 11→10 (≈9% cut)

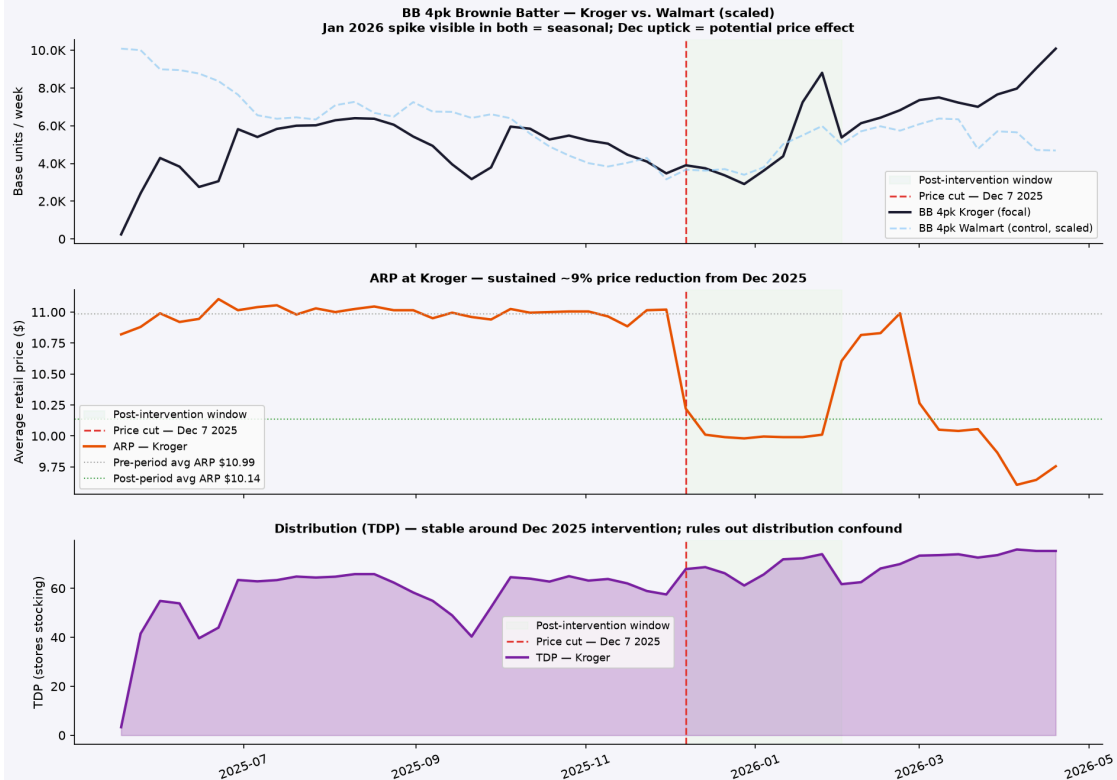


Figure 2 — Kroger BB4pk price event, December 2025. Top: Kroger (focal) vs. Walmart (control) — Jan 2026 spike appears at both retailers, confirming it is seasonal, not price-driven. Middle: Average Retail Price (ARP) trajectory — ~9% reduction sustained through Q1 2026. Bottom: TDP (distribution) stable around the intervention — rules out distribution as a confound. Net causal lift: +28.6% above BSTS baseline.

Trade spend payoff: You now know what the price-only contribution was (+4.7%) versus what the full promo package added (+28.6% net). The \$23.7pp gap is the display and feature mechanic — and it can be evaluated against the trade spend cost to determine true ROI. This is the calculation your team has been doing manually, with far less precision.

Price Scenario Modeling — Forward-Looking

What if we drop Kroger price by 5%?

Mo uses $\epsilon = -0.59$ for Kroger to estimate the demand response — and shows the confidence range around that estimate. Your trade team gets the expected unit lift and can compare it against the margin cost of the TPR before committing to the promotion.

Where should we prioritize promo spend?

Elastic accounts (Ahold $\epsilon = -1.26$) respond strongly to price moves; inelastic accounts (Walmart $\epsilon = -0.25$) do not. Mo gives you an account-level elasticity ranking so trade dollars go where they generate the most incremental volume.

Explainability & Complete Audit Trail

A finance team that has been burned before needs to know where every number came from. Mo provides a complete data lineage: from raw SPINS transactions, through every normalization and feature engineering step, to the model prediction — at any level of the hierarchy. Every number is inspectable. Every calculation is reproducible. Every forecast has a ranked list of business drivers that explains why it landed there.

SHAP Waterfall — Every Prediction Has a Business Reason

The SHapley Additive exPlanations (SHAP) waterfall below shows exactly what moved the Brownie Batter 4pk forecast at Walmart. Each bar is a business factor with a direction and a magnitude. Demand momentum, TDP trajectory, and seasonal pattern account for the majority of the prediction. **Actual: 27,317 units. Model: 28,472 units. Error: 4.2%.** If a reviewer wants to understand why that number is what it is, they can read it directly from the chart — in business language, not model vocabulary.

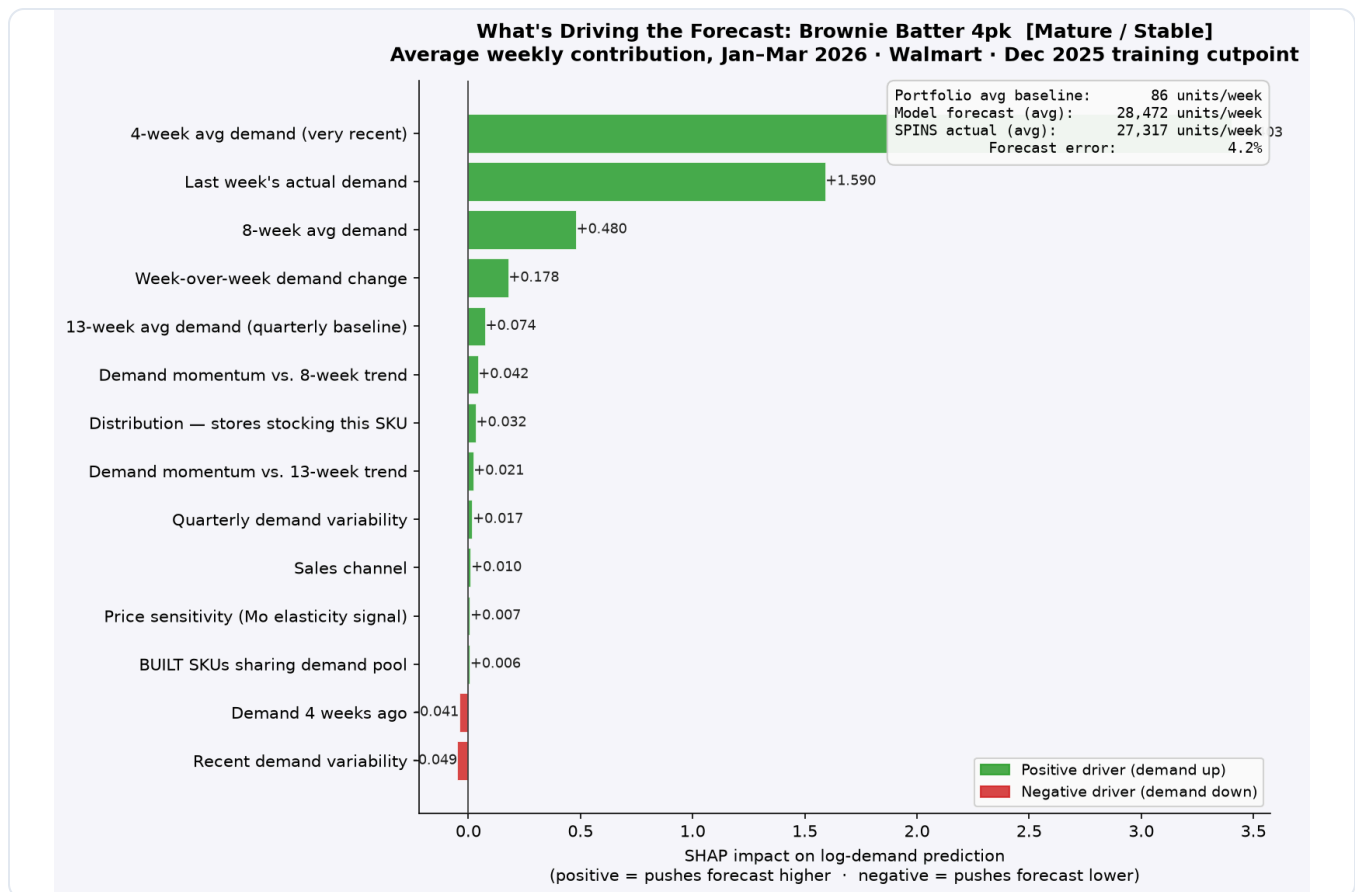


Figure 3 — SHAP waterfall: Brownie Batter 4pk at Walmart (Dec 2025 cutpoint). Base value = portfolio average demand. Each bar = one feature's contribution to the final prediction.

Data Lineage & Provenance — Audit-Ready by Design

For due diligence, valuation, or board review: Mo can document the operations performed on source data — what was kept, what was normalized, what was excluded and why — so that any reviewer can trace a forecast output all the way back to the raw SPINS transaction. This is the paper trail that normally requires an external firm and months of manual reconstruction. With Mo, it is built into every training run.

Mo Chat — Ask in Plain English

"Why is the Walmart forecast up 18% this quarter?" Mo responds with the driver decomposition — distribution change, velocity trend, seasonal pattern — grounded in live SPINS data. No spreadsheet required.

Forecast Drawer

Per-SKU, per-retailer floor/plan/ceiling for the next 13 weeks. Drill from portfolio to retailer to SKU in three clicks. Updated automatically every SPINS cycle.

Auto-Generated Briefing

Every four-week SPINS cycle regenerates this briefing automatically with updated accuracy metrics, updated forecasts, and updated elasticity. Your team sees the accuracy trend over time — not a one-time snapshot.

Growth Quality & New Product Visibility

BUILT is not a steady-state brand. Distribution expansion can make total units rise even when per-store velocity is flat — and a simple trend forecast will mistake new doors for durable demand. Mo separates these two stories every week, so your team knows whether growth is earned or distribution-dependent before you plan the next quarter around it.

TDP Decomposition — Distribution vs. Velocity

At Walmart, Brownie Batter 4pk and Cookie Dough Chunk growth has been primarily a distribution story — Total Distribution Points, or TDP (the number of stores carrying the product), rising while per-store velocity holds flat. Mo weights TDP features heavily in those series and knows not to project velocity-led acceleration that isn't happening. When velocity *does* start running ahead of distribution, Mo sees it first — and the forecast reflects it.

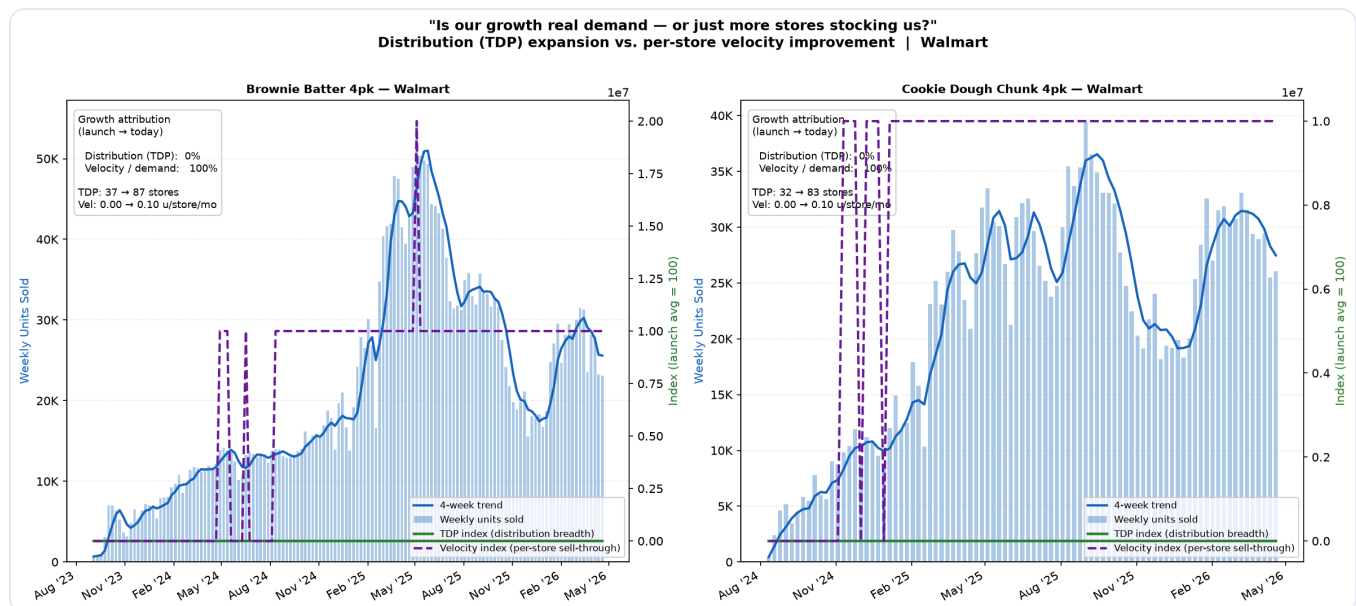


Figure 4 — Demand decomposition: BB4pk and CDC4pk at Walmart. Green = Total Distribution Points (TDP) index. Purple dashed = velocity index (sell-through per store). When TDP leads and velocity is flat, growth is a distribution story — not an organic demand story.

What Growth Decomposition Means for Your Plan

Growth Type	What Mo Sees	FP&A Implication
Distribution-led growth	Total Distribution Points (TDP) rising; velocity stable or slightly declining (normalization)	Growth is real but depends on continued shelf wins; plan conservatively if reset risk exists
Velocity-led growth	TDP stable or modest; velocity rising — consumers rebuying faster	Higher-quality demand signal; more repeatable; less dependent on new retail relationships
Promo-led volume	Units spike during promo weeks; base demand is stable or declining	~30% of BUILT portfolio is promo-linked; plan margin and inventory differently for this portion
New SKU cannibalization	New SKU velocity rising; adjacent BUILT SKU velocity declining at same retailer	Total BUILT units may hold flat while mix shifts; revenue and margin quality change with it

Context that matters: BUILT's velocity is trending down slightly as distribution expands — but the brand remains the highest-velocity product in the top-10 bar set at key accounts. Mo holds this context in the model rather than flagging velocity softness as a negative signal when it is actually normalization.

SECTION 6

What Your Finance Team Gets

Mo is a finance control layer for demand planning — not an analytics experiment. Your team gets tighter numbers, cleaner ROI reads, and a systematic way to prove out every figure in the operating plan, from raw SPINS data to the revenue call.

Revenue Planning Gets a Tighter Number

13-week retailer forecasts grounded in current SPINS — with a realistic range — replace single-point extrapolations that fail when a retailer resets or runs a promotion.

Inventory Risk Gets a Range

Floor/plan/ceiling per account gives supply chain a practical downside and upside. No more safety stock built on optimistic point forecasts.

Trade Spend Gets Accountability

Causal counterfactual shows what would have happened without the promo. Price-only lift vs. total event lift, separated and quantified, every time.

New SKU Launches Get a Guard Rail

Cannibalization pressure tracked from week 8 onward. Know whether a new flavor is growing the category or cannibalizing Brownie Batter before you commit to the full inventory build.

Price Decisions Get Evidence

Per-retailer elasticity tells you where price moves will lift units and where they won't. Spend promo dollars where the math supports it.

Analyst Capacity Gets Extended

Automated reports delivered every four weeks. Your analysts spend more time on interpretation, scenario planning, and stakeholder communication — and less on rebuilding the same SPINS pull from scratch.

The 90-Day Trust Path

Weeks	What Happens	What Your Team Sees
Weeks 1–4	First full SPINS retrain on production data; Mo forecasts run in parallel with existing process	Side-by-side comparison: Mo vs. your team's existing forecast on the same SKU x retailer series

Weeks	What Happens	What Your Team Sees
Weeks 5–8	Second retrain; accuracy trend begins; cannibalization and elasticity signals reviewed with your FP&A team	Accuracy trend visible across two training cycles; first causal event and cannibalization reviews
Weeks 9–13	Third retrain; promo calendar integration when available; elasticity validated against Q3 events	Rolling weighted forecast error tracker shows trajectory; BUILT's own baseline is the denominator, not an abstract benchmark

Inputs That Compound Accuracy Further

H2 Promo Calendar

Forward-looking promo dates convert retroactive inference to predictive modeling. Estimated +3–5pp wMAPE improvement on high-velocity, trade-heavy SKUs — equivalent to \$3–5M at \$1M per 1pp when the promotion calendar is available ahead of the event.

Your Team's Forecast Accuracy Baseline

Actual weighted forecast error on 3–5 key SKU × retailer pairs replaces the abstract benchmark with BUILT's own number. This anchors the ROI claim in your actuals, not an industry estimate.

Planned Distribution Changes

Shelf resets, new retailer launches, planogram expansions as forward-looking TDP signals. Knowing a Walmart reset in advance closes the 4–8 week SPINS reflection lag entirely.

Executive Takeaway

Mo gives your FP&A team a more precise demand signal, a causal read on price and promotion effects, SKU-level portfolio pressure visibility, and a complete audit trail — all on live SPINS data, retraining every four weeks, without additional headcount.

WHAT IS PROVEN

- 4.4% wMAPE at SKU×retailer level in Q1 2026 — tighter than BUILT's corporate total, at a harder granularity
- Per-retailer price elasticity confirmed on real events: Kroger BB4pk +28.6% total lift, +4.7% price-only
- Cannibalization pressure tracked 8-week rolling across full BUILT portfolio
- Growth decomposed: distribution-led vs. velocity-led vs. promo-led — visible every week
- Every number has a ranked business driver explanation and a data lineage your team can audit

WHAT CHANGES FOR FP&A

- Forecast accuracy becomes measurable at SKU×retailer level on live data, refreshed every four weeks
- Promo calendars and price moves become planning inputs, not explanations after the fact
- Parallel accuracy comparison can run without additional analyst load or manual model maintenance
- Within 90 days, finance has an accuracy record proven on BUILT's own data, SKUs, retailers, and demand history